Case 3

Done	
Status	Cases

ACh life cycle and mode of transmission

ACh is built from two precursors, stored in vesicles, and — unlike the monoamines — is destroyed in the cleft by enzymes rather than taken back up. Only its choline fragment is recycled, which is why the degrading enzyme, not a transporter, is the drug target.

Why this pathway matters: it defines where a drug can raise ACh levels.

1. Choline (from diet and intraneuronal stores) and acetyl coenzyme A (made from glucose in the mitochondria) meet choline acetyltransferase (ChAT) → ACh is synthesised in the cholinergic terminal.
2. The vesicular ACh transporter (VAChT) loads ACh into synaptic vesicles → ACh is available for release.
3. Released ACh acts on nicotinic and muscarinic receptors (next sections), then is hydrolysed into choline + acetate.
4. Acetylcholinesterase (AChE) = the key inactivating enzyme at the synapse; butyrylcholinesterase (BuChE), mainly glial, hydrolyses ACh that has diffused away from the synapse → action terminated too fast and too completely for the transmitter itself to be recaptured.
5. The choline transporter returns choline to the terminal → it re-enters step 1.

- Consequence of step 4–5: boosting cholinergic transmission means blocking the esterase, not blocking reuptake.
- AChE and BuChE are separate genes with different tissue distributions, so they can be inhibited separately; BuChE matters more where glia proliferate (gliosis in advanced dementia).

Synaptic vs volume transmission

- Classic view: diffuse, slow modulation — cholinergic fibres have many axonal varicosities without postsynaptic densities, release sites do not line up anatomically with receptor locations, and extracellular ACh fluctuates too slowly for localised clearance.
- Counter-view: behaviour needs faster signals than diffusion allows — cue detection in prefrontal cortex is accompanied by ACh transients on a seconds timescale.
- Working synthesis: ACh operates on two timescales at once — a slow tonic level setting the state of a region, and fast phasic release carrying discrete signals. Computational models usually implement only the slow component, as uniform modulation of all cells in the circuit.

Evidence: optogenetic stimulation of endogenous cholinergic fibres evokes fast nicotinic EPSPs in cortical and hippocampal GABAergic interneurons, restricted to specific interneuron subsets. Fast, cell-type-specific transmission therefore exists alongside volume transmission, and cholinergic action is not purely diffuse.

Nicotinic receptors: fast, calcium-permeable, self-limiting

nAChRs are ligand-gated cation channels assembled as pentamers of α and β subunits. Which subunits are in the ring decides agonist affinity, calcium permeability and how fast the receptor shuts itself off — so subunit composition, not the transmitter, carries the functional specificity.

- Two classes, defined originally by binding: receptors binding agonists with high affinity but not α-bungarotoxin (the heteromeric "nAChRs": α2–α6 with β2–β4), and receptors binding agonists weakly but α-bungarotoxin strongly (the αBgtx class: α7–α10, usually homomeric).
- Binding sites sit at subunit interfaces and have a principal and a complementary component.
    - Heteromeric: the principal component is on the α subunit, the complementary one on β → two ACh sites per receptor.
    - Homomeric α7: every subunit contributes both components → five sites per receptor.

    - α5 and β3 carry neither component: they are auxiliary subunits that do not bind ACh but change permeability, desensitisation and localisation of the receptor they join.
- Four conformational states: resting, open, and two desensitised closed states (recovering over milliseconds or minutes) that are refractory to activation yet bind agonist with very high affinity. Ligands anywhere on the receptor shift the equilibrium between states → sustained agonist exposure silences rather than drives the receptor.
- Calcium is the main currency: nAChRs pass Na^+ and Ca^{2+}, and homomeric α7 is as calcium-permeable as NMDA receptors but needs no depolarisation to let calcium in. Calcium entry triggers transmitter release presynaptically, depolarises postsynaptically, and acts as a second messenger for survival and gene expression.
- Desensitisation as protection: α7's rapid desensitisation limits calcium influx and prevents excitotoxicity. Mutating the leucine residue in the second transmembrane domain that drives desensitisation yields a slowly desensitising, higher-affinity receptor — and, in vivo, excess calcium entry, apoptosis and seizure susceptibility.

The two subtypes that matter for cognition

	α4β2	α7
Agonist sensitivity	high affinity, responds to low (basal) ACh	low affinity, also activated by the precursor choline
Kinetics	slower	fast activation, fast desensitisation
Calcium	moderate permeability	high permeability
Typical placement	postsynaptic; on thalamocortical terminals; regulates dopamine release in nucleus accumbens	presynaptic (facilitates transmitter release) and postsynaptic/perisynaptic in prefrontal cortex and hippocampus
Functional read	tonic activation by ambient ACh; target of nicotine, hence tobacco reinforcement	phasic, coincidence-type signal; shapes cortical cognitive circuits

- Stoichiometry is not fixed: the same α4 and β2 subunits can assemble in different ratios, giving populations that differ in activation sensitivity and desensitisation rate. Chronic nicotine exposure shifts the mix toward the high-affinity form — a receptor-level form of plasticity in long-term smokers.
- Positive allosteric modulators (PAMs) bind away from the ACh site and increase channel opening frequency only when ACh is present → amplification that follows endogenous activity instead of replacing it.

Muscarinic receptors: slow, bidirectional, position-defined

mAChRs are G-protein-coupled, so they act slowly and can excite or inhibit depending on coupling. The family splits cleanly by G protein, and that split maps onto where the receptor sits: the excitatory ones are postsynaptic, the inhibitory ones are presynaptic brakes.

- M1, M3, M5 couple to Gq/11 → phospholipase C → excitatory postsynaptic effects: depolarisation and suppression of spike-frequency accommodation (the cell keeps firing instead of slowing down).
- M2, M4 couple to Gi/o → inhibition of adenylyl cyclase → inhibitory effects, and they sit presynaptically.
    - On cholinergic terminals = autoreceptors: ACh that builds up in the cleft shuts off its own further release.
    - On GABA and glutamate terminals = heteroreceptors: ACh diffusing away from its synapse blocks release of the other transmitter there. This is the mechanism behind cholinergic suppression of intrinsic cortical transmission (next section).
- Distribution worth holding: M1 dominates cortex and hippocampus and is the memory-relevant subtype; M4 is densest in striatum, associated with dopaminergic regulation; M5 is sparse but sits on dopamine neurons of the substantia nigra and VTA.
- Blocking muscarinic receptors (scopolamine, atropine) is the standard way to remove cholinergic signalling experimentally and the source of the anticholinergic cognitive side-effect profile.

Why selective muscarinic drugs are hard

- The ACh binding site (orthosteric site) is highly conserved across M1–M5, so an orthosteric agonist cannot be made subtype-selective → inadvertent action on peripheral M2 (cardiovascular) and M3 (gastrointestinal, visual) receptors produces the side effects that limit cholinergic drugs.
- Allosteric sites are far less conserved between subtypes → they are the route to selectivity.
    - Allosteric agonist: produces a response on its own and also enhances ACh's effect.
    - PAM: no effect by itself, only facilitates the response to the orthosteric agonist → it rides on endogenous activity.
    - Allosteric antagonist: lowers affinity or efficacy for the orthosteric ligand.

Evidence: xanomeline, an M1/M4-preferring partial agonist, improved cognition and psychotic symptoms in schizophrenia but was abandoned for peripheral cholinergic side effects — the clinical demonstration that efficacy was never the problem, selectivity was.

Where ACh comes from

Two architectures with different jobs: projection nuclei that broadcast ACh to distant targets, and local interneurons that set a high tonic level inside one structure. The architecture predicts the signal — a broadcast system can set a global state, a local one works by pausing.

- Basal forebrain projection system (medial septum, diagonal band of Broca, nucleus basalis magnocellularis / nucleus basalis of Meynert).
    - Medial septum and vertical limb of the diagonal band → hippocampus and entorhinal cortex; this is the memory arm, and entorhinal cortex gets among the densest cholinergic innervation of any cortical structure.
    - Nucleus basalis and horizontal limb → neocortex and amygdala; this is the attention arm, and the one that degenerates early in Alzheimer disease.
- Brainstem projection system (pedunculopontine and laterodorsal tegmental nuclei) → thalamus and basal ganglia, with a minor contribution to cortex and

a feedback input to basal forebrain. Its projection into the VTA triggers dopamine release, linking cholinergic drive to the addictive properties of nicotine.

- Local interneurons: striatum and nucleus accumbens are densely innervated by intrinsic cholinergic cells that do not project outside the structure and fire tonically; a smaller interneuron population exists in neocortex and hippocampus.
- Regulatory feedback into basal forebrain comes from a restricted set of sources — glutamatergic input from medial/ventral prefrontal cortex (the route for top-down recruitment of cholinergic drive), GABAergic input from accumbens and hippocampus, and ascending aminergic input — i.e. the cholinergic system broadcasts widely but is itself steered by frontal cortex.
- Functional asymmetry to carry forward: in projection systems the behavioural signal is a rise in ACh; in the striatum ACh tone is already high, so the signal is a pause in interneuron firing at behaviourally relevant stimuli.

The core circuit effect: extrinsic in, intrinsic out

The single mechanism that most cholinergic cognitive effects reduce to: ACh strengthens afferent (extrinsic, sensory) input while suppressing intrinsic (recurrent, feedback) transmission. The network is pushed from "recall what I already know" toward "take in what is out there", which is simultaneously a signal-to-noise improvement and a bias toward encoding.

The problem it solves: a recurrent network that is always listening to itself overwrites new input with previously stored patterns, so there must be a switch between processing the world and processing one's own representations.

1. High ACh acts via nicotinic receptors on afferent terminals (α4β2 on thalamocortical terminals; nicotinic potentiation in hippocampal dentate gyrus and CA1) → more glutamate released from the incoming sensory pathway = the signal is amplified.
2. High ACh acts via muscarinic heteroreceptors on intrinsic/feedback terminals (piriform cortex, hippocampal CA3) → presynaptic inhibition of glutamate release at recurrent synapses only, sparing afferent synapses = previously stored associations are kept from being retrieved.

3. ACh depolarises pyramidal cells and suppresses spike-frequency accommodation (M1) → the cells that do receive afferent input stay responsive.
4. The combination raises the ratio of feed-forward to feedback drive = "signal-to-noise" in cortical networks, and in sensory cortex sharpens receptive fields by reducing spatial spread of activity.
5. In parallel, ACh lowers the threshold for inducing LTP at the active synapses → whatever is processed under high ACh is also encoded.

- Why not keep the network in this state permanently: low ACh restores intrinsic transmission, which is needed to consolidate and to retrieve stored patterns; and gating which inputs dominate allows the same stimulus to be treated differently depending on context.
- The cortical microcircuit version: nicotinic receptors increase excitability of dendrite-targeting interneurons while muscarinic receptors reduce GABA release from soma-targeting (parvalbumin) interneurons. Net effect — thalamic feed-forward input drives pyramidal firing more easily, while intracortical input onto distal dendrites is damped.
- Breakdown of the switch is pathological: models in which encoding and retrieval are no longer separated show runaway strengthening of synapses, with degeneration and amnesia resembling Alzheimer disease.

Evidence: in piriform (olfactory) cortex, cholinergic agonists selectively suppress transmission at feedback synapses but not at afferent synapses from the lateral olfactory tract — the finding that established the afferent/intrinsic dissociation as a property of the synapse type rather than of the cell. Simulating it in a network model of piriform cortex showed that without this suppression, recurrent input corrupts the encoding of a new stimulus resembling earlier ones.

Attention

Attention is treated here as *effective processing*: the ability of neural tissue to update its activation pattern in response to external stimuli enough to meet task demands and resolve ambiguity. The extrinsic-in/intrinsic-out mechanism is exactly what effective processing requires, so attention is its behavioural readout.

- Two timescales in prefrontal cortex, matching the two modes of transmission.
    - Fast transients (seconds) accompany detection of a cue; they appear in medial prefrontal cortex but not motor cortex, even though both receive cholinergic projections → release is targeted, not global.
    - Slow changes (minutes) in baseline ACh across trials → these track *attentional effort*, i.e. demand, rather than performance as such: ACh efflux rises with how hard the task taxes attention even when accuracy does not improve.
- The signal is specific to detection, not to responding: failed cue detections do not produce the transient, and habituated operant responses are not accompanied by it.
- Receptor division of labour mirrors the core mechanism: amplification of afferent signals depends on nicotinic receptors, while suppression of intrinsic transmission depends on muscarinic receptors. Accordingly, blocking muscarinic but not nicotinic receptors abolishes the attentional modulation of V1 neurons.
- Where the control comes from: prefrontal cortex projects back to the basal forebrain, so attentional effort can be recruited top-down — cholinergic drive is both the mechanism of attention and an output of frontal control.
- Alternative reading of the same data: rather than raising signal-to-noise, ACh may gate bottom-up versus top-down processing — signalling how far the current context can be trusted, so that top-down expectations are allowed in when sensory evidence is ambiguous. The two accounts differ in whether background (intrinsic) activity is noise to be removed or useful contextual disambiguation.

Paradigm: the sustained attention task (SAT)

Rats continuously monitor a house light and must press one lever when it is on, another when it is off; signals appear at irregular intervals so attention cannot be timed. Cortical ACh efflux rises sharply while animals perform the task, but only slightly during matched control tasks equated for movement and novelty — which is how ACh release was tied to attention rather than to arousal or motor activity.

Evidence: selective cholinergic deafferentation of cortex (immunotoxin into nucleus basalis) produces deficits in stimulus detection that extensive retraining cannot reverse, and residual performance across animals tracks how much of the cholinergic projection survived. Lesions of prefrontal cholinergic input also increase susceptibility to distractors — consistent with loss of the intrinsic-suppression half of the mechanism.

Working memory

Working memory holds information actively during a delay. ACh matters for it specifically when the material is novel, because the cellular mechanism it switches on — persistent spiking — is what carries a representation that has no strengthened synapses to fall back on.

- Novel vs familiar is the organising distinction.
    - Familiar stimuli, small stimulus sets: handled by prefrontal and parietal cortex, where synaptic connections are already strengthened → little cholinergic input needed, and muscarinic blockade spares simple short-term tasks such as digit span.
    - Novel, trial-unique stimuli: handled by parahippocampal/entorhinal structures acting as a buffer → cholinergic modulation is required.
- Cholinergic depletion of prefrontal or perirhinal cortex, and systemic muscarinic blockade, impair delayed-match-to-sample performance; the M1 subtype is implicated, since an M1 antagonist produces dose-dependent impairment.

Mechanism: cholinergically gated persistent spiking

The problem it solves: a representation must survive after the stimulus has gone, without relying on pre-existing synaptic strength.

1. A medial entorhinal neuron normally spikes only while a depolarising current lasts → nothing persists after the stimulus.
2. ACh binds muscarinic receptors on that neuron and opens a calcium-sensitive non-specific cation current (I_CAN) = the switch that makes persistence possible.

3. Afferent glutamatergic input during the stimulus depolarises the cell and lets calcium in.
4. Calcium activates I_CAN → further depolarisation → more spiking → more calcium entry: a self-sustaining loop = the maintenance state.
5. Spiking therefore continues after the depolarising stimulus ends, holding the item through the delay; without ACh the loop never starts and the cell returns to silence.

- Model confirmation: simulating entorhinal pyramidal cells with this muscarinic cation current reproduces sustained firing after a transient input, and removing the current degrades sustained spiking during the simulated delay.
- Why gating matters rather than simply switching it on: the maintained representation must survive distractors. Cholinergic modulation selects which stimuli enter the maintenance state, which is why scopolamine impairs tasks containing distractors (n-back) more than undistracted ones.
- Converging line: novelty — new environments and new objects — itself raises hippocampal ACh, so the system is tuned to admit exactly the information for which no stored attractor yet exists.

The same logic in episodic memory

- Encoding quality is limited by processing quality, so the attention mechanism directly predicts an encoding role: scopolamine given before a word list impairs later free recall, but the same drug after encoding does not → the cholinergic contribution is to encoding, not storage or retrieval.
- The second contribution is plasticity gating: ACh lowers the LTP threshold (cholinergic agonists make a single stimulation burst, normally ineffective, sufficient to induce LTP in CA1) and boosts theta, whose phases separate encoding from retrieval.
- Reconciling the negative findings: lesion studies that find spared memory typically use over-trained animals with familiar stimuli and low attentional demand — exactly the condition in which the cholinergic mechanism is not needed. Deficits reappear when the stimuli are novel.

Schizophrenia: a cholinergic account of the cognitive symptoms

The cognitive disturbances of schizophrenia — reduced attention and working memory — are the symptoms the cholinergic mechanisms above would produce if they failed. The nicotinic evidence converges on α7, the muscarinic evidence on M1 and M4.

Nicotinic side

- Clinical entry point: smoking is far more prevalent in schizophrenia than in the general population, suggesting self-medication with a nicotinic agonist. Patients also show much less of the smoking-induced upregulation of high-affinity nAChRs seen in controls → nicotine's effect on this receptor class is blunted, which would require heavier smoking for the same effect.
- Receptor findings: α7 immunoreactivity and α-bungarotoxin binding are reduced in hippocampus, thalamic reticular nucleus and cingulate/frontal cortex, whereas the abundant heteromeric receptors (α4, β2) look largely unchanged → the deficit is specific to the homomeric class.
- Genetics: the chromosomal locus containing the α7 gene is linked to the sensory gating deficit found in affected families, promoter polymorphisms of the α7 gene are associated with both the illness and the gating deficit, and copy number variations at that locus raise risk.
- Functional meaning: the P50 auditory evoked potential gating deficit — failure to suppress the response to a repeated stimulus — is transiently reversed by nicotine, tying α7 to the filtering of redundant sensory input.
- Mechanistic link to cognition: α7 receptors in prefrontal cortex modulate glutamatergic signalling through NMDA receptors, the circuit supporting working memory; deleting α4β2 in medial prefrontal cortex impairs visual attention, and re-expressing the receptor there restores it.

Muscarinic side

- Postmortem and imaging studies find reduced muscarinic binding across cortical and subcortical regions, resolving to lower M1 in cortex and lower M4

in hippocampus; the M1 reduction is specific to schizophrenia rather than common to psychiatric illness.

- Importantly, it is not uniform: a minority of patients carry a marked cortical M1 reduction while the rest resemble controls → a biochemically defined subgroup inside the syndrome, and a candidate for targeted rather than syndrome-wide treatment.
- Direction of causation is supported pharmacologically: muscarinic antagonists produce psychosis-like symptoms in healthy people and worsen symptoms in patients.
- Therapeutic logic that follows: M1 agonism for cognitive deficits, M4 agonism for positive symptoms (M4 regulates VTA dopamine neurons and so dampens mesolimbic dopamine release) — the two halves of xanomeline's profile, now pursued with allosteric ligands to escape the selectivity problem.
- Caveat on the nicotinic findings: heavy smoking and antipsychotic treatment both alter receptor expression, so receptor differences in patients are confounded unless controlled.

Alzheimer disease and cholinesterase inhibition

The cholinergic hypothesis of amnesia: early memory decline reflects loss of the basal forebrain cholinergic projection while its postsynaptic targets are still intact, so raising ACh at those surviving targets restores part of the lost function. This is symptomatic treatment — it replaces a signal, it does not stop the degeneration.

The treatment logic, and the reason it has an expiry date:

1. Plaques and tangles accumulate and damage the cholinergic projection neurons of the nucleus basalis → cholinergic innervation of hippocampus and neocortex is lost, while cortical and hippocampal target neurons with their receptors survive = a presynaptic deficiency with an intact postsynaptic apparatus.
2. Blocking AChE prevents the breakdown of the ACh that is still released → ACh accumulates in the cleft and reaches the surviving receptors = the lost signal is partly restored (the same logic as levodopa restoring function of a degenerating dopamine projection).

3. Clinically this more often retains the current level of memory function and slows decline than it improves memory.
4. As the disease progresses, neocortical and hippocampal neurons themselves die → the receptor targets are lost, there is nothing left for the accumulated ACh to act on, and cholinesterase inhibitors lose effectiveness.

The three inhibitors, by what distinguishes them

	Mechanism beyond AChE block	Consequence
Donepezil	reversible, long-acting, selective for AChE only	easy dosing; peripheral AChE inhibition in the gut gives mostly transient GI side effects
Rivastigmine	pseudoirreversible; inhibits BuChE as well, with some preference for AChE in cortex and hippocampus	glial BuChE inhibition adds a second route to raising ACh, which matters more in later disease when gliosis raises BuChE activity; more GI effects orally, reduced by transdermal delivery
Galantamine	AChE inhibition plus positive allosteric modulation of nicotinic receptors	the ACh gained by esterase block is amplified at nicotinic receptors — theoretically synergistic, not shown to be a clinical advantage

Receptor-level pathology

- Overall nAChR numbers fall in neocortex, hippocampus and thalamus in the absence of a general neuronal loss, driven by loss of α4-containing receptors → the loss of high-affinity agonist binding. The relative sparing of β2 is unexplained.
- β-amyloid interacts directly with both receptor classes — non-competitively inhibiting α4β2 and binding α7 with very high affinity — offering a route from plaque pathology to impaired cognition and, since both receptor classes support neuronal survival, to premature neuronal death.

Other pathologies that locate function to receptor subtypes

- Autosomal dominant nocturnal frontal lobe epilepsy: mutations in the α4 or β2 subunit genes, clustered in the channel-forming transmembrane domain, increase sensitivity to ACh → easier synchronisation of thalamocortical oscillations and seizures. The clearest demonstration that a single nAChR subunit controls cortical excitability.
- Parkinson disease: α6- and α4-containing receptors on nigrostriatal dopaminergic terminals control dopamine release and are selectively lost, while cortical α4 and α7 loss parallels the cognitive impairment — a reminder that unselective nicotinic treatment acts on many subtypes at once, which is why clinical trials of nicotine have been equivocal.